GBENGA SAOLA

Data & AI Engineer · MSc Artificial Intelligence

Open to Data, AI and Engineering roles · United Kingdom

saolagbengaifeoluwa@gmail.com
07780 250588
ife-gsaola.github.io
github.com/ife-gsaola

PROFILE

MSc Artificial Intelligence graduate with three years of professional data experience in operations analytics and data management, built on advanced SQL, data validation and automated reporting. I design and build real-time AI systems end to end: streaming speech recognition resolved with phonetic matching and sentence embeddings, a native C++ video engine, LLM application pipelines, and a multi-tenant data platform. Recently upskilled in Azure, AWS, Databricks and Apache Kafka.

SELECTED PROJECTS

Auto Verses: real-time speech-to-entity resolution

2025 – now

- Turns a live speech stream into exact references as they are spoken: streaming ASR (Web Speech or local Whisper) with interim-result handling, so results show before the sentence ends.
- Built a phonetic matcher for recognition errors (custom sound keys and edit distance over joined word windows): 27 of 28 misheard names resolved.
- Semantic search with a quantised MiniLM sentence-embedding model running offline over 62,000 rows; context tracking and voice commands; 35 automated tests.

Stack: TypeScript, Whisper, Web Speech API, MiniLM, Transformers.js, Electron, NDI

AuraCast Studio: real-time video and audio production

2025 – now

- Desktop live-production system (camera switching, graphics, mixing, streaming), used weekly on Windows and Mac.
- Native C++ streaming and recording engine up to 4K with FFmpeg (LGPL) and a Direct3D 11 path on Windows; real-time audio engine with no locks or allocation on the audio thread.
- Phones as wireless cameras via native Swift and Kotlin apps; NDI and YouTube Live output.

Stack: C++, FFmpeg, DSP, TypeScript, React, Electron, WebRTC, Swift, Kotlin

ApplyPilot: LLM pipeline for job applications

2025 – now

- Collects jobs from public APIs and careers platforms (Greenhouse, Lever, Ashby); rewrites the CV per job as reviewable diffs, with guardrails so it never invents experience.
- Generates cover letters and supporting statements; a Manifest V3 Chrome extension fills forms and submits only on the user's click; two-step sign-in.

Stack: React, TypeScript, Supabase, LLM APIs, Chrome extension, LaTeX

Sanctuary: multi-tenant SaaS data platform

2025 – now

- Isolated tenants with Postgres row-level security (500+ policies), 53 Deno serverless functions, and a WhatsApp Cloud API inbox that parses free-form messages into structured data.

Stack: Postgres, Supabase, Deno, TypeScript, React, Vercel

Data and ML projects

github.com/ife-gsaola

- Inventory data warehouse: an ETL from staging to EDW layers built with SQL stored procedures.
- Hugging Face language model deployed on an AWS SageMaker endpoint behind Lambda, with a React front end.
- CNN food-image recogniser with text-to-speech output (Keras, Streamlit, Flask); ML coursework in credit risk, forecasting, clustering and recommenders; statistics in R.

EXPERIENCE

Operations Data Analyst Nov 2020 – Oct 2023
SecureID Limited, Nigeria

- Built automated dashboards that replaced manual Excel reporting across production, quality and inventory, giving management near-real-time operational visibility.
- Designed and optimised SQL queries to reconcile operational and financial datasets, improving confidence in metrics reported at senior level.
- Analysed performance trends and presented data-driven recommendations that supported cost control and capacity planning.
- Identified process improvements that contributed to faster order fulfilment and better resource use.

Data Management Executive Sep 2020 – Mar 2021
Ipsos, Nigeria

- Engineered SQL validation and cleansing logic that automated manual quality checks, increasing processing speed and dataset reliability.
- Standardised data-integrity checks in SQL, SPSS and Excel to find and resolve anomalies such as straight-lining and outliers.
- Translated business requirements into deliverables for sales forecasting and strategic decisions.

Professional Development & Independent Projects Aug 2025 – Apr 2026
Self-directed

- Upskilled in cloud and data engineering platforms: Azure, AWS, Databricks and Apache Kafka.
- Google Project Management and IBM Database Administrator certificates (Coursera, 2025); data modelling and warehousing projects.

Embedded Software Training Apr 2024 – Aug 2025
TekTrain, online

- Programmed ESP32 and Arduino microcontrollers in C++, writing, compiling and flashing firmware to hardware; integrated temperature, weather and humidity sensors.

Carer Apr 2026 – now
Care UK, Birmingham

- Person-centred care in a high-accountability setting; accurate, contemporaneous records kept to information-governance and confidentiality standards.

EDUCATION

MSc Artificial Intelligence 2025
University of Wolverhampton
Reinforcement learning, neural networks, supervised and unsupervised learning; simulation-based RL agents in Unity (C#)

BSc Civil Engineering 2019
Osun State University

SKILLS

AI & ML	Python, reinforcement learning, neural networks, model training and evaluation, TensorFlow/Keras, scikit-learn, Hugging Face, Whisper, sentence embeddings, LLM APIs
Data	SQL (advanced), BigQuery, Databricks, Apache Kafka, data warehousing, pipeline design, data validation and quality controls, KPI reporting, Power BI, Qlik Sense, SPSS, R, Excel (advanced)
Cloud & platforms	AWS (SageMaker, Lambda), Azure, Supabase, Postgres (row-level security), Vercel
Software	TypeScript, React, Node, Electron, C++, C#, Swift, Kotlin, Unity, ESP32/Arduino, Git

CERTIFICATES

Google Project Management (Coursera, 2025) · IBM Database Administrator (Coursera, 2025) · Data Science Boot Camp (Udemy, 2020) · Excel for Data Analysis (Simplilearn, 2019)